OTDR



Category: **Telecom & Networking** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: OTDR

What it is used for

An optical time-domain reflectometer (OTDR) tests fibre-optic links by sending light pulses into a fibre and analysing the returned signal. It helps locate breaks, splices, connectors, and loss events along the fibre route.

Main parts

- OTDR main unit and display
- Optical output port and connector adapter
- Launch fibre and receive fibre when required
- Test leads and approved fibre connectors
- Battery or power adapter
- Cleaning tools and optical inspection accessories

Before you start

1. Read the OTDR manual and identify the fibre type, wavelength, and test limits.
2. Confirm that the fibre link is disconnected from live transmitters and is safe to test.
3. Inspect the OTDR port, test leads, launch fibre, and connectors for damage or contamination.
4. Clean and inspect fibre connectors using approved fibre-optic cleaning tools.
5. Select the correct test wavelength and refractive index for the fibre under test.
6. Set the range, pulse width, averaging time, and launch fibre arrangement appropriate to the link.

How to use

1. Connect the launch fibre to the OTDR and the fibre link according to the test plan.
2. Connect the receive fibre at the far end when required to evaluate the final connector or link end.
3. Verify that all connections are secure and that the test path is correctly identified.
4. Start the OTDR acquisition and allow the instrument to collect the required trace.
5. Review the trace for events such as connectors, splices, bends, breaks, and attenuation changes.
6. Check event distance, reflectance, and loss values against the acceptance criteria.
7. Repeat the test at the specified wavelength or direction when required by the test plan.
8. Save the trace and record fibre identification, wavelength, settings, and test results.
9. Disconnect the test leads carefully and replace protective caps on clean connectors.
10. Switch off the OTDR and store the unit, launch fibre, and accessories in their protective case.

⚠ Safety

- Never look directly into a fibre end or OTDR optical port during testing.
- Verify that the fibre is disconnected from active optical transmitters before connecting test equipment.
- Use only compatible optical connectors and never force a connector into the port.
- Keep fibre ends and glass fragments away from skin and eyes; collect scraps in a designated sharps container.
- Clean connectors with approved tools and avoid touching the polished end face.
- Follow site procedures for working near live telecom equipment and elevated or confined routes.

Care & storage

- Inspect and clean optical connectors before and after each test.
- Protect the OTDR port and fibre ends with clean dust caps when not in use.
- Inspect launch fibres, test leads, adapters, and cables for damage or tight bends.
- Keep software, calibration status, battery condition, and test records up to date.

Common mistakes to avoid

- Testing a live fibre can expose the user to invisible laser radiation and damage the instrument.
- Dirty connectors can create false loss events and unreliable readings.
- Using an incorrect refractive index or pulse width can produce inaccurate event distances or resolution.
- Skipping the launch fibre can prevent reliable measurement of the near-end connector.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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